Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: **B. Tech** Semester: **III**

Name of the Course: **Fundamentals of AI ML** Course Code: **TCS 364**

Time: **3 Hours** Maximum Marks: 100

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1 Attempt any Two (20 Marks)

(a) Explain the AI agent concept with the help of suitable diagram. Explain the working principles of simple reflex agents and model-based agents with real-world examples. — CO1

(b) Consider a graph. — CO1

Start node: S (OR node). Edges and nodes:

- S –(2)→ A (AND node)
  - A –(2)→ A1 (leaf) with heuristic h(A1)=3
  - A –(2)→ A2 (leaf) with heuristic h(A2)=4
- S –(3)→ B (OR node)
  - B –(2)→ B1 (leaf) with heuristic h(B1)=6
  - B –(4)→ B2 (leaf) with heuristic h(B2)=2

Use the AO* algorithm to find the optimal solution graph from S (i.e., the least-cost AND–OR solution).

(c) Explain the evolution, challenges and significant application areas of AI.

Q2 Attempt any Two (20 Marks)

(a) — CO2

Customer	Age	Income	BuysComputer
C1	Young	High	No
C2	Young	High	No
C3	Middle	High	Yes
C4	Senior	Medium	Yes
C5	Senior	Low	Yes
C6	Senior	Low	No
C7	Middle	Low	Yes
C8	Young	Medium	No
C9	Young	Low	Yes
C10	Senior	Medium	Yes

A company wants to predict whether a customer will Buy a Computer (Yes/No) based on two features: Age and Income. The dataset is shown above. Using the Naïve Bayes classifier, predict whether a new customer with: 'Age=Young, Income=Medium will Buy a Computer or not'.

(b) Differentiate the propositional and predicate logic with suitable examples. — CO2

Given the following statements, convert them into predicate logic:

- "All humans are mortal."
- "Socrates is a human."

Then, infer whether Socrates is mortal.

(c) Explain how knowledge-based AI systems use inference mechanisms to derive conclusions from a knowledge base. — CO2

Q3 Attempt any Two (20 Marks)

(a) Explain the simple linear regression in detail. A company wants to predict Sales (Y) based on Advertising Budget for TV (X1) and Radio (X2). The dataset is: — CO4

Observation	X1 (TV)	X2 (Radio)	Y (Sales)
1	1	2	4
2	2	1	5
3	3	4	7
4	4	3	10

(a) Fit a multiple linear regression equation for the above data:

Y=a+b1X1+b2X2 using least squares.

(b) Predict sales when TV = 5 and Radio = 4.

(b) Discuss various performance measures used by supervised and unsupervised machine learning models with their formula, preferences according to use and importance. — CO3

CO3

(c) You are given the dataset with two features (X1, X2) and class labels:

Point	X1	X2	Class
P1	1	2	A
P2	2	3	A
P3	3	3	B
P4	6	5	B
P5	7	7	B

A new data point Q(3,4) needs to be classified using the K-Nearest Neighbor (KNN) algorithm.

(a) Compute the Euclidean distance from Q to each data point (P1–P5).

(b) Classify Q for k = 3 and k=4.

(c) If Distance-Weighted KNN is used (weight = 1/distance), what will be the predicted class?

Q4 Attempt any Two (20 Marks)

(a) How a binary classification problem is solved using Support Vector Machine (SVM)? CO5
(a) Explain the role of the hyperplane and margin in classification. CO5
(b) How can we draw a simple 2D example with support vectors and show how the decision boundary is chosen. CO4

(b) You are given 10 training examples. Each example has attributes Temperature, Humidity, Windy and target class Play (Yes/No). Build a decision tree using entropy / information gain.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Q4-(c)

(c) Differentiate any three
i. Bagging and boosting
ii. Bias and Variance
iii. Overfitting and Under fitting
iv. Classification and Regression

Q5 Attempt any Two (20 Marks)

(a) Explain the process of evaluating a classification model using training–testing data split. Derive and explain the formulas for accuracy, precision, recall, and F1-score, and illustrate their computation using a sample confusion matrix. CO5

(b) Discuss the importance of ROC curve and AUC in evaluating machine learning models. Plot a sample ROC curve using given True Positive Rate (TPR) and False Positive Rate (FPR) values, and explain how AUC helps in model comparison. CO5

(c) Explain the K-Means clustering algorithm step by step with a suitable example.
Perform one iteration of K-Means on the following data points with k = 2 and initial centroids (2, 4) and (6, 8):
Data points = {(2,5), (3,4), (5,7), (6,9)}. Show all distance calculations and updated centroids. CO6